Thermodynamic-limit inference from quantum measurements with discrete diffusion

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Abstract

Quantum many-body phenomena often require access to large system sizes beyond current experiments and simulations. Here we introduce Diffushadow, a discrete diffusion framework that learns transferable quantum measurement statistics across system sizes through full-attention iterative denoising. Trained on small-system classical-shadow data, Diffushadow generates measurement records for larger unseen systems without retraining. Starting from exact data for systems of at most 14 spins, recursive scale extrapolation extends generation to 96 spins and supports thermodynamic-limit estimates of correlations and structure factors in transverse-field Ising, frustrated $\boldsymbol{J_1}$–$\boldsymbol{J_2}$, and axial next-nearest-neighbour Ising chains. In two dimensions, a model trained on lattices of at most 16 spins preserves selected spatial correlations on unseen 49-spin lattices. The same denoising formulation reconstructs partially observed Rydberg occupation records with near-Bayes-optimal per-site accuracy without task-specific retraining. Together, these results establish cross-scale quantum measurement generation as a route to many-body inference beyond directly accessible system sizes.

Keywords: quantum many-body systems, discrete diffusion models, thermodynamic-limit inference, finite-size scaling, attention analysis

Quantum many-body systems exhibit collective phenomena, including magnetic ordering and quantum phase transitions, whose characterization often requires understanding the thermodynamic limit [1, 2]. Accessing this limit remains challenging, as the Hilbert-space dimension grows exponentially with system size, constraining exact numerical calculations [3]. Although advances in programmable quantum simulators and quantum computers have enabled the preparation of increasingly complex quantum states [4, 5], acquiring sufficiently precise measurement statistics can require substantial experimental resources, while accurate numerical references remain difficult to obtain for many large systems. These challenges motivate approaches that exploit measurements from accessible finite systems to infer physical properties at larger scales and support thermodynamic-limit estimation.

In recent years, machine-learning approaches have increasingly been used to characterize quantum many-body systems, including property prediction [6, 7], phase identification [8, 9], and quantum-state reconstruction [10, 11]. Classical shadows provide a measurement-efficient representation of randomized quantum measurements from which multiple observables can be estimated [12]. Generative approaches go further by learning the measurement distribution itself. ShadowGPT models Hamiltonian-conditioned classical-shadow distributions with an autoregressive Transformer [13], while M-Diffushadow introduced multimodal discrete diffusion for measurement generation and reconstruction in the TFI model [14]. Conditional generative models have also demonstrated proof-of-concept size extrapolation in larger Rydberg chains [15]. However, a limitation arises for autoregressive measurement models, whose fixed causal ordering makes bidirectional use of observations distributed throughout a configuration less direct. This becomes particularly relevant for two-dimensional lattices, where no single linear ordering is intrinsic to the spatial geometry, and for incomplete records, where observed and missing outcomes can be interspersed across the system. Existing generative models of quantum measurements have primarily focused on prescribed system sizes, with only limited demonstrations of extrapolation to nearby larger systems, leaving transfer from small systems to substantially larger unseen sizes largely unexplored.

In this context, scalability in machine learning for quantum many-body systems can refer to different goals, including processing measurements from large systems, exploiting locality to reduce sample complexity, or generalizing across states and Hamiltonian parameters [9, 15, 16]. Here, we focus on cross-scale generative transfer by investigating whether a model trained only on randomized measurements from small systems can generate complete measurement configurations at larger unseen sizes. Such generation would extend the range of system sizes available for finite-size scaling toward the thermodynamic limit, particularly near critical regimes where finite-size effects become important [17].

In this study, we introduce Diffushadow, a discrete diffusion framework [18–20] for cross-scale quantum measurement generation. The framework transfers measurement statistics learned from small systems to larger unseen sizes and, through recursive scale extrapolation, progressively extends the system sizes available for finite-size analysis. Full-attention iterative denoising with spatial positional encoding enables this scale transfer without imposing a fixed causal generation order. We

show that this cross-scale generative capability can be recursively extended to support thermodynamic-limit inference of selected observables across the one-dimensional TFI, frustrated J_1–J_2, and ANNNI models. We further demonstrate direct cross-scale generation on unseen two-dimensional transverse-field Ising lattices. Beyond scale transfer, the same denoising formulation also reconstructs incomplete Rydberg occupation records without task-specific retraining. Attention analysis also provides complementary evidence that the model's spatial information aggregation exhibits systematic correspondence with physical correlation structures.

Results

Diffushadow

Diffushadow is a discrete diffusion framework for cross-scale generation of quantum measurement records (Fig. 1). Each classical-shadow snapshot [12, 21] contains a sequence of randomly selected Pauli measurement bases $\mathbf{P}$ and corresponding joint outcomes $\mathbf{b}$, from which multiple physical observables can be estimated [22]. Rather than predicting individual observables directly, Diffushadow learns the conditional measurement distribution $p(\mathbf{b} \mid \lambda, \mathbf{P})$, where λ denotes the Hamiltonian control parameter. Measurement bases and outcomes are represented as distinct site-resolved tokens, allowing the same model architecture to process sequences of different system sizes.

Diffushadow learns this distribution through masked discrete denoising with full self-attention. Unlike causal attention, full attention allows Diffushadow to predict masked outcomes using information from both preceding and subsequent sites, conditioned on the Hamiltonian parameter, measurement bases and available outcomes (Fig. 1b). This distinction is illustrated by a causal-mask control in which fixing the measurement outcome of one qubit primarily affects predictions at later sequence positions under causal attention, whereas full-attention Diffushadow propagates the change to predictions on both sides of the selected site (Extended Data Fig. 1).

Measurement generation proceeds through iterative unmasking. Starting from a fully masked outcome sequence, the model predicts the conditional outcome distributions at all remaining masked sites. At each iteration, the site with the highest predictive confidence is selected, its outcome is sampled and fixed, and the remaining predictions are recomputed using the updated measurement context. The resulting generation order adapts to the model's evolving predictions rather than following a predetermined sequence. Compared with one-shot prediction of all outcomes, iterative decoding substantially reduces errors in the tested TFI observables (Extended Data Table 1).

To support cross-scale generation, Diffushadow combines mixed-size training with spatial positional encodings for site positions and two-dimensional lattice coordinates, allowing application to unseen sizes without architectural changes. For further scale extension, recursive scale extrapolation incorporates generated larger-system measurements into successive training rounds alongside retained exact small-system data (Fig. 1c). The same denoising formulation reconstructs incomplete records by keeping observed outcomes fixed while iteratively filling missing positions.

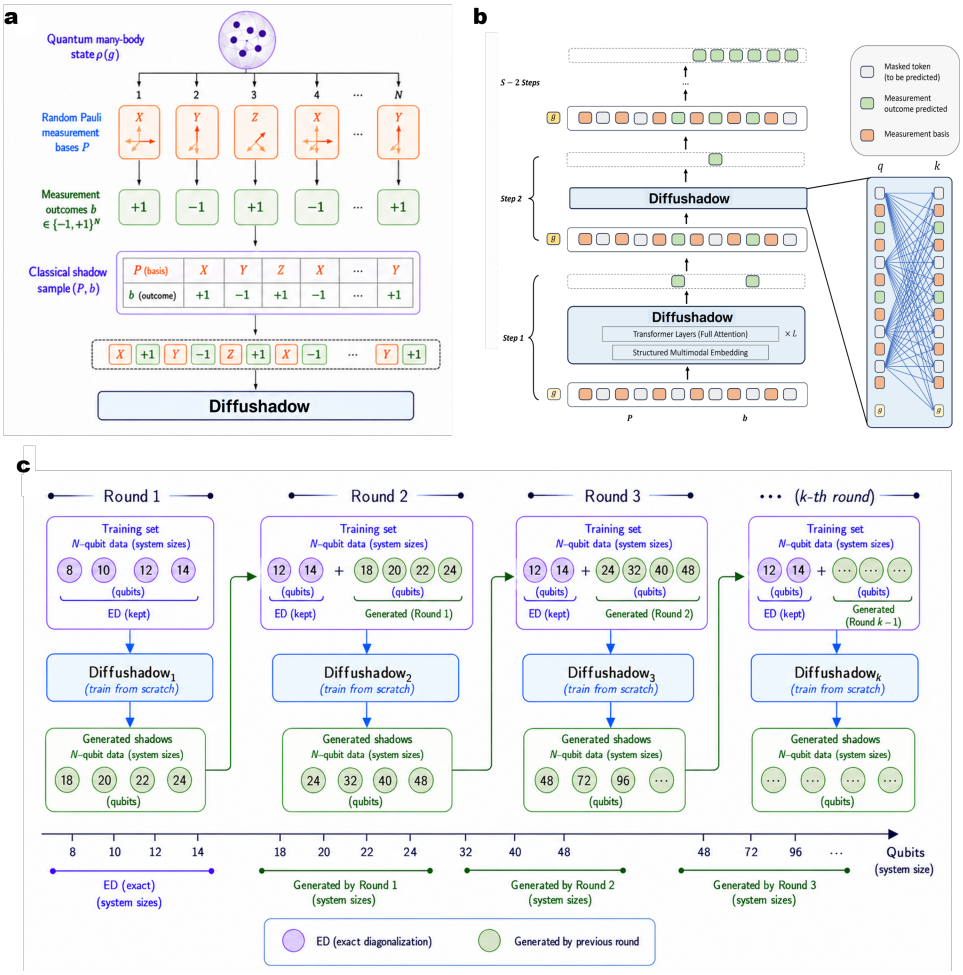


Fig. 1 Overview of the Diffushadow framework for cross-scale quantum measurement generation. **a** Classical-shadow representation of a quantum many-body state. Each qubit is measured in a randomly selected Pauli basis, and the corresponding measurement outcomes are recorded. The basis sequence P and outcome sequence b together form a classical-shadow snapshot, which is represented as a discrete token sequence for Diffushadow. **b** Diffusion-based generation of quantum measurement snapshots. Starting from a fully masked measurement sequence, Diffushadow iteratively denoises the configuration and predicts masked measurement outcomes. Full attention allows the model to aggregate information from both measurement-basis and measurement-outcome tokens across the entire sequence, enabling collective refinement without imposing an autoregressive ordering. **c** Recursive scale extrapolation. Diffushadow is first trained on exact small-system classical-shadow data obtained from exact diagonalization, and is then used to generate shadow data for larger systems beyond the original training sizes. The generated data are incorporated into subsequent training rounds together with the retained exact small-system data, progressively expanding the accessible system size. Purple circles denote exact ED data, green circles denote generated data, and braces group the system sizes according to their data source.

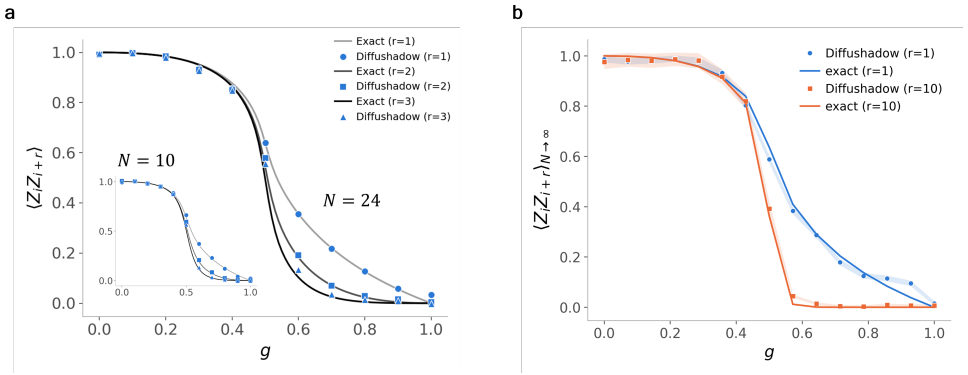


Fig. 2 Cross-scale generation and thermodynamic-limit inference in the transverse-field Ising model. a, Longitudinal spin correlations ($\langle Z_i Z_{i+r} \rangle$) for $r = 1, 2, 3$ at the unseen system size $N = 24$, with the training size $N = 10$ shown in the inset. Solid curves show exact diagonalization results, and markers show estimates from Diffushadow-generated classical-shadow measurements. **b**, Thermodynamic-limit correlations for $r = 1$ and $r = 10$, obtained by linear extrapolation in $1/N$ of observables estimated from recursively generated measurements. Solid curves show exact thermodynamic-limit results, and markers show Diffushadow estimates. Shaded regions indicate pointwise 95% bootstrap confidence intervals.

Cross-scale generation and thermodynamic-limit inference in the TFI model

To assess cross-scale measurement generation, we consider the one-dimensional transverse-field Ising (TFI) model with periodic boundary conditions, described by $H = -(1-g)\sum_i Z_i Z_{i+1} - g\sum_i X_i$, where g controls the competition between ferromagnetic interactions and transverse-field quantum fluctuations. In the thermodynamic limit, the critical point $g_c = 0.5$ separates ferromagnetic and paramagnetic phases. Longitudinal spin correlations $\langle Z_i Z_{i+r} \rangle$ characterize the spatial correlations of the ground state, remaining extended in the ordered regime and decaying with separation in the paramagnetic regime. Their dependence on both g and r also provides a probe of the measurement statistics generated by Diffushadow.

Cross-scale measurement generation in the TFI model

We first evaluate measurement generation at the training size $N = 10$. As shown in the inset of Fig. 2a, correlations at $r = 1, 2, 3$ estimated from Diffushadow-generated classical shadows closely follow exact diagonalization (ED) results across the scanned values of g, including the rapid suppression of longitudinal correlations near the critical region. Comparisons of the ground-state energy and selected spin correlations with the autoregressive ShadowGPT baseline [13] are reported in Extended Data Table 1, where Diffushadow achieves lower mean squared errors across the reported observables and coupling values.

For direct cross-scale generation, Diffushadow is trained on mixed-size measurements from $N = 8, 10, 12, 14$ and evaluated on the unseen $N = 24$ chain without retraining. Figure 2a shows that the generated measurements reproduce both the field

dependence and spatial separation of the longitudinal spin correlations. At weak transverse fields, Diffushadow recovers the near-unity correlations at all three separations, characteristic of the ferromagnetic regime. Near the critical region, the generated estimates follow the rapid suppression of correlations and the increasing differences between $r = 1, 2, 3$. In the paramagnetic regime, Diffushadow distinguishes the more persistent nearest-neighbour correlations from the faster decay of correlations at larger separations, rather than reproducing only their overall decrease with g. Deviations from ED remain visible near the critical region and at selected larger field strengths, particularly for the nearest-neighbour correlation. The simultaneous recovery of field-dependent and distance-dependent correlation behaviour at $N = 24$ demonstrates that the learned measurement statistics remain transferable beyond the system sizes represented in training.

Thermodynamic-limit inference from recursively generated measurements

To extend the system sizes available for finite-size inference, we employ recursive scale extrapolation (Fig. 1c). Starting from exact measurements for systems with at most $N = 14$ spins, each subsequent round trains a newly initialized model on larger-system measurements generated by the preceding model, together with retained exact small-system data. Three recursive rounds extend generation to $N = 96$ without introducing additional exact large-system measurements. Observable estimates at $N = 24$ remain largely consistent across different recursive rounds (Extended Data Fig. 2), indicating limited drift at the tested scale.

The recursively generated measurement ensembles provide finite-size estimates of the longitudinal correlations $\langle Z_i Z_{i+r} \rangle$. For each field strength g and separation r, we perform an empirical linear extrapolation in inverse system size [23, 24], using $C_{ZZ,N}(r, g) = C_{ZZ,\infty}(r, g) + a(r, g)/N$. The intercept at $1/N = 0$ gives the thermodynamic-limit estimate, which is compared with the exact reference in Fig. 2b. Pointwise 95% bootstrap confidence intervals quantify the sampling uncertainty propagated through the finite-size fits.

Figure 2b compares the thermodynamic-limit correlations inferred from recursively generated Diffushadow measurements with exact TFI results at $r = 1$ and $r = 10$. The extrapolated estimates closely follow the reference profiles across the scanned transverse fields, reproducing strong correlations at both separations in the ferromagnetic regime and their rapid suppression near the critical region. Importantly, Diffushadow captures the contrasting behaviour of the two correlation distances in the paramagnetic regime. The longer-range $r = 10$ correlation rapidly approaches zero, whereas the nearest-neighbour correlation remains appreciable and decreases more gradually with increasing field. This agreement shows that the recursively generated measurement ensembles retain sufficient spatial correlation information to distinguish short- and longer-range behaviour after extrapolation to the thermodynamic limit. Although deviations remain at selected field strengths, particularly for $r = 1$ in the paramagnetic regime, Diffushadow reproduces the principal features of both exact thermodynamic-limit correlation profiles.

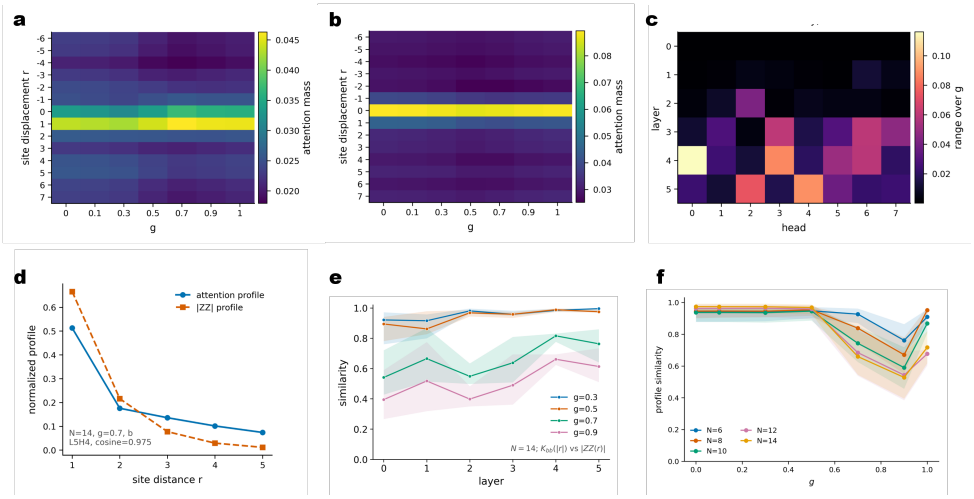


Fig. 3 Spatial attention patterns and alignment with spin correlations in the transverse-field Ising model. Attention is evaluated with all measurement-outcome tokens masked. Panels a–e use $N = 14$. **a,b,** Attention kernels from masked outcome queries to outcome tokens, $K_{bb}(r, g)$ (a), and measurement-basis tokens, $K_{bP}(r, g)$ (b), as functions of signed periodic site displacement r and transverse-field parameter g. Colours indicate attention mass averaged over query sites, measurement-basis configurations, heads and layers. **c,** Variation in the attention-weighted mean site distance across g for each layer–head pair. Colour indicates the range $\Delta D^{(\ell,h)} = \max_g D^{(\ell,h)}(g) - \min_g D^{(\ell,h)}(g)$. **d,** Absolute-distance attention kernel $K_{bb}(|r|)$ and exact absolute correlation profile $|\langle Z_i Z_{i+r}\rangle|$ for head L5H4 at $g = 0.7$, selected for the highest cosine similarity at this setting. Each profile is normalized to unit sum over the displayed distances. **e,** Median cosine similarity across heads within each layer between outcome-directed attention kernels and exact absolute ZZ correlation profiles, shown at $g = 0.3, 0.5, 0.7$ and 0.9. **f,** Median cosine similarity across all layer–head pairs as a function of g for $N = 6, 8, 10, 12$ and 14. Shading denotes the interquartile range across heads within each layer in e and across all layer–head pairs in f. Layer and head indices are zero-based.

Correspondence between attention patterns and physical correlations

We examine the spatial organization of attention in Diffushadow and its correspondence with spin correlations in the TFI model [25]. All measurement-outcome tokens are masked, while the transverse-field parameter and measurement bases remain available. The outcome-directed kernel $K_{bb}(r, g)$ and basis-directed kernel $K_{bP}(r, g)$ exhibit distinct spatial patterns (Fig. 3a,b). Outcome-directed attention is concentrated at the nearest-neighbour displacement $r = 1$, with a longer-range component that is more pronounced in the ferromagnetic regime and weakens as the transverse field increases into the paramagnetic regime. This coupling-dependent localization is consistent with the decreasing spatial extent of longitudinal spin correlations. In contrast, basis-directed attention remains strongly concentrated at the same site ($r = 0$), indicating a persistent preference for the local measurement basis.

The spatial response to the transverse field differs substantially across attention heads and layers. We quantify this dependence through the variation $\Delta D^{(\ell,h)}$ of the attention-weighted mean site distance across g (Fig. 3c). Heads in earlier layers generally exhibit small variations, whereas several heads in deeper layers show considerably

greater changes in their spatial attention range. This heterogeneous sensitivity suggests that different heads contribute differently to the field-dependent organization of spatial information, rather than adjusting their attention ranges uniformly across the network.

We further assess whether these attention patterns correspond to physical correlations by comparing the absolute-distance outcome-directed kernel $K_{bb}(|r|)$ with the exact absolute correlation profile $|\langle Z_i Z_{i+r}\rangle|$. At $N = 14$ and $g = 0.7$, head L5H4, selected for the highest cosine similarity at this setting, exhibits a similar distance-dependent decay to the correlation profile, with a similarity of 0.975 (Fig. 3d). The layer-wise analysis extends this comparison to all eight heads in each layer (Fig. 3e). At $g = 0.3$ and 0.5, the median similarity remains high throughout the network, whereas at $g = 0.7$ and 0.9, alignment becomes weaker and more heterogeneous across layers and heads. Some deeper layers retain relatively strong correspondence, but similarity does not increase monotonically with depth. Although parameter sensitivity and correlation-profile similarity measure distinct properties, their heterogeneous distributions are consistent with differentiated roles of attention heads and layers in processing spatial information.

This regime-dependent correspondence persists across system sizes $N = 6, 8, 10, 12$ and 14 (Fig. 3f). The median similarity over all layer–head pairs remains high in the ferromagnetic regime and at the critical coupling $g_c = 0.5$, but decreases in the paramagnetic regime, reaching lower values near $g = 0.9$. The alignment with physical correlation profiles therefore extends beyond the selected head and system size, while exhibiting systematic dependence on the physical regime and substantial variation within the network.

Cross-scale generation across different Hamiltonians and lattice geometries

We further evaluate Diffushadow on quantum systems with competing interactions and different lattice geometries. For the frustrated J_1–J_2 and axial next-nearest-neighbour Ising (ANNNI) chains, we use recursively generated measurements to estimate thermodynamic-limit observables. For the two-dimensional TFI model, we examine direct cross-scale generation on larger unseen periodic lattices without retraining.

Frustrated and competing-interaction spin chains

The antiferromagnetic J_1–J_2 chain features competing nearest- and next-nearest-neighbour exchange interactions, with a transition from the gapless to the dimerized regime at $J_{2c}/J_1 \simeq 0.241$ and the Majumdar–Ghosh (MG) point at $J_2/J_1 = 0.5$ (Fig. 4b) [26, 27]. We probe these interactions through the spin-dot correlations $\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+r}\rangle$ at $r = 1, 2$. The ANNNI chain combines ferromagnetic nearest-neighbour and competing next-nearest-neighbour Ising interactions in a transverse field (Fig. 4d) [28, 29]. At $\kappa = 0.4$, below the zero-field multiphase point $\kappa = 0.5$, we examine $S_Z(\pi)$ and $S_Z(\pi/2)$, which probe staggered and period-four correlation components, respectively.

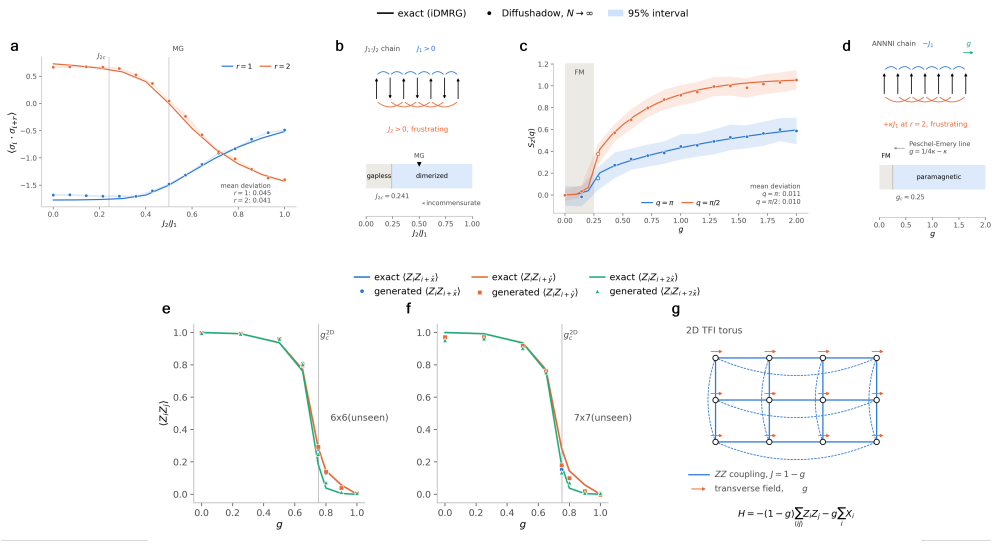


Fig. 4 Thermodynamic-limit inference in frustrated spin chains and cross-scale generation on two-dimensional lattices. a, Thermodynamic-limit spin correlations $\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+r} \rangle$ for $r = 1, 2$ in the frustrated J_1–J_2 chain. Vertical lines mark the critical coupling $J_{2c}/J_1 \approx 0.241$ and the Majumdar–Ghosh (MG) point at $J_2/J_1 = 0.5$. **b,** Interaction schematic and phase diagram of the J_1–J_2 chain, indicating the gapless and dimerized regimes, MG point and onset of incommensurate correlations. **c,** Thermodynamic-limit longitudinal structure factors $S_Z(\pi)$ and $S_Z(\pi/2)$ in the ANNNI chain at $\kappa = 0.4$. Grey shading indicates the ferromagnetic regime. In a,c, solid curves show iDMRG references, and markers show Diffushadow estimates linearly extrapolated in $1/N$. Shading denotes pointwise 95% bootstrap confidence intervals. Annotated mean absolute deviations exclude the open markers near the ANNNI transition in c. **d,** Interaction schematic and phase diagram of the ANNNI chain, indicating the ferromagnetic and paramagnetic regimes. **e,f,** Site-averaged ZZ correlations on unseen 6×6 (e) and 7×7 (f) periodic TFI lattices, generated by a model trained on lattices of at most 16 spins. Solid curves show SSE quantum Monte Carlo references; markers show Diffushadow estimates. Blue circles, orange squares and green triangles denote horizontal nearest-neighbour $\langle Z_i Z_{i+\hat{x}} \rangle$, vertical nearest-neighbour $\langle Z_i Z_{i+\hat{y}} \rangle$ and horizontal next-nearest-neighbour $\langle Z_i Z_{i+2\hat{x}} \rangle$ correlations, respectively. Vertical lines mark $g_c^{\mathrm{2D}} \approx 0.753$. **g,** Periodic square-lattice geometry with nearest-neighbour ZZ coupling $J = 1 - g$ (blue bonds) and transverse field g (orange arrows). Dashed bonds indicate periodic connections.

For both chains, we evaluate generated measurements at the training size $N = 12$ and unseen size $N = 24$ (Extended Data Fig. 3), then extrapolate observables from recursively generated larger-size ensembles linearly in $1/N$. The intercepts are compared with independent infinite-system density-matrix renormalization group (iDMRG) references; pointwise 95% bootstrap confidence intervals propagate finite-record sampling uncertainty through observable estimation and extrapolation.

In the J_1–J_2 chain, the spin-dot observable combines XX, YY and ZZ correlations, testing whether Diffushadow preserves statistics across Pauli components rather than only the longitudinal ZZ correlations of the TFI model (Fig. 4a). At weak frustration, nearest-neighbour correlations are strongly negative and next-nearest-neighbour correlations positive. As J_2/J_1 increases, Diffushadow captures the weakening of the former and the sign reversal of the latter near the MG point. At stronger frustration,

it also reproduces the increasing magnitude of next-nearest-neighbour antiferromagnetic correlations relative to nearest-neighbour correlations. This redistribution across the gapless and dimerized regimes reflects the changing competition between exchange interactions. Mean absolute deviations from iDMRG are 0.045 ($r = 1$) and 0.041 ($r = 2$).

The ANNNI chain tests momentum-resolved structure factors, which combine longitudinal correlations across multiple spatial separations (Fig. 4c). At $\kappa = 0.4$, both $S_Z(\pi)$ and $S_Z(\pi/2)$ are small in the low-field ferromagnetic regime, where the dominant ordering wavevector is $q = 0$. As the transverse field increases, Diffushadow reproduces the rapid rise and broad plateau of $S_Z(\pi/2)$, alongside the more gradual growth of $S_Z(\pi)$. Diffushadow also preserves the separation between the two profiles, including the greater spectral weight at $q = \pi/2$. Because the two wavevectors weight spatial separations differently, their simultaneous agreement with iDMRG indicates that the generated ensembles retain distinct components of the spatial correlation structure, not merely its overall magnitude. Mean absolute deviations from iDMRG are 0.011 and 0.010 for $S_Z(\pi)$ and $S_Z(\pi/2)$, respectively, excluding open markers near the transition where reference-grid interpolation is less reliable.

Cross-scale generation on two-dimensional lattices

We evaluate direct cross-scale generation in the two-dimensional TFI model on periodic square lattices, where nearest-neighbour ZZ interactions couple spins along both spatial directions and periodic bonds connect opposite lattice boundaries (Fig. 4g). Unlike one-dimensional chains, these two-dimensional neighbour relationships cannot be represented uniformly as local adjacency under a fixed linear ordering. We train a separate Diffushadow model on classical-shadow measurements from mixed-size two-dimensional lattices containing at most 16 spins, using two-dimensional rotary positional embeddings to encode lattice coordinates. Without retraining, the model generates measurement records for unseen 6×6 and 7×7 lattices, containing 36 and 49 spins, respectively. We compare site-averaged horizontal and vertical nearest-neighbour ZZ correlations, together with horizontal next-nearest-neighbour correlations, against stochastic series expansion quantum Monte Carlo (SSE QMC) references.

Figures 4e,f show that Diffushadow closely reproduces the SSE QMC correlations on both unseen lattice sizes. The periodic square geometry gives equivalent local environments along the horizontal and vertical directions, for which the exact nearest-neighbour correlations are equal by lattice symmetry. Diffushadow recovers this near-equivalence in its generated measurements while accurately following the field dependence of both directional correlations. The model also reproduces the distinct behaviour of horizontal next-nearest-neighbour correlations, which are more strongly suppressed than nearest-neighbour correlations as the transverse field increases. The agreement across different spatial directions and separations indicates that the generated measurements preserve the tested directional symmetry and distance-dependent correlation structure beyond the training sizes.

Near the two-dimensional critical coupling $g_c^{2D} \simeq 0.753$, Diffushadow captures the rapid suppression of longitudinal correlations on both unseen lattice sizes, including the different rates of decay of nearest- and next-nearest-neighbour correlations (Fig. 4e,f). The SSE QMC references also exhibit visible finite-size differences between the 6×6 and 7×7 lattices in this region. Diffushadow follows these size-dependent variations rather than reproducing identical correlation profiles at the two sizes, although deviations from the references remain near the transition. Notably, this agreement extends to the unseen 7×7 lattice, containing 49 spins compared with at most 16 in training. The results show that direct cross-scale generation preserves the tested spatial correlations and their finite-size dependence across substantially larger two-dimensional lattices, without retraining or recursive scale extrapolation.

Reconstruction of incomplete Rydberg measurement records

Measurements from programmable atom arrays can be incomplete owing to atom loss or readout failure, reducing the number of records usable for multisite observables. In Rydberg systems, spatial occupation correlations and blockade constraints provide information about missing outcomes through the remaining observations. Diffushadow exploits this conditional information by holding observed outcomes fixed while reconstructing missing positions through iterative denoising, without task-specific retraining. We evaluate this capability on an open Rydberg chain of $N = 32$ atoms at blockade radius $R_b/a = 2.35$, using held-out occupation snapshots sampled from DMRG ground states. The model is trained on chains with $N = 16, 24, 32$, and 30% of the sites are randomly masked at test time. Conditional sampling is used to reconstruct measurement ensembles, whereas deterministic decoding predicts individual missing outcomes (Fig. 5a).

We first assess whether the reconstructed ensembles preserve collective spatial correlations. At $\delta/\Omega = 5.0$, Diffushadow recovers the characteristic Z_3 peak of the connected density–density structure factor at $q = 2\pi/3$, with an amplitude of 0.536 compared with the DMRG reference value of 0.539 (Fig. 5b). In contrast, an unconditioned-fill control, which inserts occupations from independently generated records conditioned only on detuning, produces a substantially reduced peak of 0.309. This difference shows the importance of conditioning on the observed occupations of each individual record. By incorporating these observations, Diffushadow recovers the spatially organized correlations that are substantially weakened when missing sites are filled without regard to their measurement context.

Conditional completion also preserves local physical consistency between observed and reconstructed sites. Across the tested detunings, the fraction of doubly excited nearest-neighbour pairs spanning these sites remains below 0.02%, compared with up to 3.7% for the unconditioned-fill control (Supplementary Information). This low violation rate is achieved without explicitly enforcing blockade constraints during decoding, indicating that the learned conditional measurement statistics retain the local restrictions associated with Rydberg interactions. The reconstruction further supports multisite observable estimation. At $\delta/\Omega = 5.0$, Diffushadow reproduces intact-data estimates of two- and eight-site occupation correlators, evaluated at sites separated according to the Z_3 period, to within approximately 0.002. Meanwhile, the

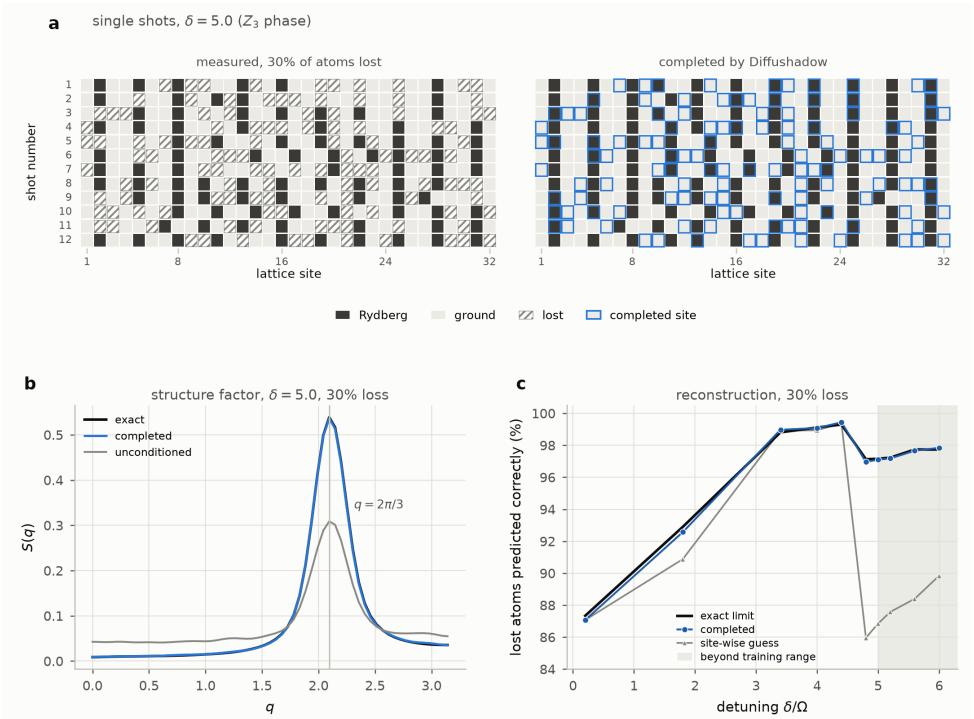


Fig. 5 Reconstruction of incomplete Rydberg measurement records. Results for an open chain of $N = 32$ atoms at blockade radius $R_b/a = 2.35$, using held-out occupation snapshots sampled from DMRG ground states. Missing readouts are simulated by randomly masking 30% of the sites in each snapshot. Diffushadow reconstructs the masked outcomes by conditional denoising while keeping observed outcomes fixed. **a,** Twelve partially observed snapshots at $\delta/\Omega = 5.0$ (left) and their conditional completions (right). Dark and light cells denote Rydberg and ground-state occupations, respectively; hatched cells indicate missing readouts, and blue outlines identify completed sites. **b,** Connected density–density structure factor $S(q)$ at the same detuning. Black, blue and grey curves show the DMRG reference, conditionally completed ensemble and unconditioned-fill control, respectively. The control fills missing sites using model samples conditioned only on detuning. The vertical line marks the Z_3 correlation peak at $q = 2\pi/3$. Panels **a,b** use conditional sampling. **c,** Prediction accuracy at masked sites as a function of δ/Ω, using deterministic decoding (blue circles). The black curve shows the Bayes-optimal per-site accuracy computed from conditional marginals of the DMRG reference state. Grey triangles show a site-wise baseline that predicts the more probable occupation from each site's unconditional density. The shaded region indicates detunings beyond the training range, $\delta/\Omega > 5.0$. Within each comparison, the same missing-site masks are used for Diffushadow and the corresponding references or controls.

average numbers of usable records increase from 467 and 28, respectively, to all 1000 test records (Supplementary Information). Thus, conditional reconstruction enables substantially more incomplete records to contribute to multisite correlation estimates while maintaining close agreement with the corresponding intact-data observables.

For individual-outcome prediction, we compare deterministic Diffushadow decoding with the Bayes-optimal expected per-site accuracy obtained from conditional occupation marginals of the reference DMRG matrix product state (Fig. 5c). These marginals account for the same observed occupations available to Diffushadow and

define the optimal expected accuracy for predicting each missing site. We also evaluate an unconditional site-density baseline that predicts the most probable occupation at each position without conditioning on the observed record. At $\delta/\Omega = 5.0$, Diffushadow achieves 97.1% accuracy, closely approaching the Bayes-optimal reference of 97.2% and substantially exceeding the baseline accuracy of 86.9%. The model remains close to the Bayes-optimal reference across the tested detunings, including $\delta/\Omega = 5.2, 5.6, 6.0$ beyond the training range $\delta/\Omega \leq 5.0$. This agreement indicates that Diffushadow effectively exploits the conditional information in partially observed records for missing-site prediction, with the learned conditional statistics remaining useful even at unseen detunings.

Discussion

Diffushadow demonstrates that quantum measurement statistics learned from small systems can be transferred to larger unseen sizes and used for many-body inference. By modeling the joint distribution of measurement outcomes rather than predicting individual observables directly, the framework generates reusable measurement ensembles for estimating different correlations and structure factors. Recursive scale extrapolation extends generation from exact data for systems of at most $N = 14$ spins to $N = 96$, supporting thermodynamic-limit estimates through finite-size extrapolation across three one-dimensional Hamiltonians. Direct generation on larger two-dimensional lattices further demonstrates the transfer of spatial correlation statistics across system sizes. Diffushadow therefore extends the finite-size measurement information available for physical inference without requiring additional exact large-system measurements.

Attention analysis provides a complementary view of the spatial information processing within Diffushadow. In the TFI model, attention patterns exhibit a regime-dependent correspondence with spin correlations, while differences across heads and layers suggest differentiated roles in organizing spatial information. These observations provide an internal signature of physically structured attention, although their causal contribution to generative accuracy and cross-scale transfer remains to be established [30].

The Rydberg results demonstrate a further use of the learned conditional measurement distribution. Without task-specific retraining, the same denoising formulation reconstructs missing outcomes while retaining observed measurements. Conditional sampling preserves collective correlations and largely maintains local blockade constraints, whereas deterministic decoding approaches the Bayes-optimal expected per-site accuracy. These complementary capabilities distinguish ensemble-level reconstruction from individual-outcome prediction and support the use of incomplete records for multisite observable estimation [31].

Several limitations define the scope of these findings. Validation covers selected Hamiltonians, observables and lattice geometries, with attention analysis restricted to the one-dimensional TFI model. Thermodynamic-limit estimates rely on an empirical linear extrapolation in $1/N$, whose accuracy may be limited near criticality. The reported bootstrap intervals quantify finite-sample uncertainty but do not capture systematic generation bias, errors accumulated through recursive training or uncertainty

in the extrapolation form. Furthermore, all measurement records are numerically generated, and Rydberg missing readouts are simulated by random masking. Validation on experimental measurements with realistic readout noise and missingness mechanisms remains necessary [31–33]. Near-Bayes-optimal per-site accuracy also does not imply optimal reconstruction of the joint missing-site configuration. Extending these investigations to experimental data and more complex spatial correlations will help establish the broader reliability of cross-scale quantum measurement inference.

Methods

Measurement datasets and numerical references

All measurement datasets were generated numerically from ground states of the models described in Results. For the one-dimensional TFI, J_1–J_2 and ANNNI chains and the two-dimensional TFI lattices, small-system ground states were obtained by exact diagonalization with periodic boundary conditions, using a matrix-free Lanczos solver [34] for the two-dimensional systems. Classical-shadow snapshots [12] were generated by independently and uniformly selecting a measurement basis from $\{X, Y, Z\}$ at each site and sampling the joint outcomes according to the Born rule through sequential projective measurements. Each snapshot contains the measurement bases and binary outcomes for all sites. For the Rydberg chains, we used open boundary conditions and obtained ground states as finite matrix product states (MPSs) using density-matrix renormalization group (DMRG) [35]. Occupation records were drawn from the MPSs by direct sampling [36] in the fixed Z basis, with independently sampled held-out records used to evaluate reconstruction. Numerical references were computed directly from exact-diagonalization ground states for accessible finite systems, using infinite-system DMRG (iDMRG) [37] for the thermodynamic-limit J_1–J_2 and ANNNI observables, and using stochastic series expansion quantum Monte Carlo [38] for the larger two-dimensional TFI lattices. The Rydberg reference MPSs additionally provided density and correlation observables and conditional occupation marginals for evaluating reconstruction against the Bayes-optimal expected per-site accuracy.

Diffushadow architecture

Diffushadow is implemented as a full-attention Transformer for discrete classical-shadow measurement sequences [12, 39]. Each measurement snapshot is represented as an interleaved token sequence containing the Hamiltonian control parameter, the local measurement bases, and the corresponding measurement outcomes. For a system of N spins, the input sequence is written as

$$\mathbf{x} = [\boldsymbol{\lambda}, P_1, b_1, P_2, b_2, \ldots, P_N, b_N], \tag{1}$$

where $\boldsymbol{\lambda}$ denotes the scalar control parameter used in the corresponding experiment, $P_i \in \{X, Y, Z\}$ is the Pauli measurement basis on site i, and b_i is the measurement outcome.

The scalar control parameter is mapped to the model hidden dimension through a linear projection. The measurement-basis tokens and measurement-outcome tokens are mapped through separate embedding layers, as they encode distinct types of information. For outcome tokens, an additional mask category is included to represent corrupted measurement outcomes during diffusion training. To distinguish the three types of tokens in the sequence, namely the control parameter, measurement bases, and measurement outcomes, we add a learnable type embedding to each token representation. The resulting embedded sequence is then passed to a stack of Transformer blocks. Unless otherwise specified, Diffushadow consists of $L = 6$ Transformer blocks with $H = 8$ attention heads per block, a hidden dimension of $d_{\mathrm{model}} = 256$, and a feed-forward dimension of $d_{\mathrm{ff}} = 1024$. The same architectural configuration is used for the TFI, J_1–J_2, and ANNNI experiments.

Each Transformer block consists of multi-head full self-attention followed by a feed-forward network, with residual connections and layer normalization. Unlike causal autoregressive models, Diffushadow applies full attention over the entire sequence, allowing each measurement-outcome token to attend to the control parameter, all measurement bases, and all other available outcomes. To encode positional information across variable system sizes, Diffushadow applies rotary positional embeddings (RoPE) to the query and key representations in each attention layer [40]. RoPE injects site-order information by applying position-dependent rotations to the query and key vectors, so that attention weights depend not only on token content but also on relative token positions. For two-dimensional lattices, the positional representation is extended to encode the two lattice coordinates; implementation details are provided in the Supplementary Methods. The attention output is then projected back to the hidden dimension and passed through a feed-forward network with GELU activation.

The final output layer is a linear classifier applied only to the measurement-outcome positions. For each site i, the model predicts a two-dimensional distribution over the binary measurement outcome,

$$p_\theta(b_i \mid \boldsymbol{\lambda}, \mathbf{P}, \mathbf{b}_t), \tag{2}$$

where $\mathbf{b}_t$ denotes the current partially masked measurement-outcome sequence.

Training Diffushadow

Diffushadow is trained with a stochastic masking objective on quantum measurement records. For each snapshot, the conditioning information consists of the Hamiltonian control parameter $\boldsymbol{\lambda}$ and the measurement-basis sequence $\mathbf{P}$, while the complete measurement-outcome sequence is denoted by $\mathbf{b}_0$. The same objective is applied to Rydberg occupation records, with all measurement-basis tokens fixed to Z.

During training, only the measurement outcomes are corrupted, while $\boldsymbol{\lambda}$ and $\mathbf{P}$ remain unchanged. For each sample, a masking probability p_{mask} is drawn uniformly from $[\epsilon, 1]$, where $\epsilon > 0$. Each outcome token is then independently replaced with [MASK] with probability p_{mask}, yielding the partially masked sequence $\mathbf{b}_t$. Sampling different masking probabilities exposes the model to contexts containing different numbers of observed outcomes.

For mixed-size training, each mini-batch contains records from a single system size, ensuring a common sequence length within each optimization step. Samples are shuffled within each size group, and the resulting mini-batches are shuffled across groups at each epoch.

The model is optimized to recover the original outcomes at masked positions using the Hamiltonian parameter, measurement bases and available outcomes. For a mini-batch $\mathcal{B}$ containing B snapshots of system size N, the denoising loss is

$$\mathcal{L}_{\mathcal{B}}(\theta) = -\frac{1}{n_{\mathrm{mask}}} \sum_{s=1}^{B} \sum_{i=1}^{N} m_{s,i} \log p_\theta \left(\mathbf{b}_{0,s}[i] \mid \boldsymbol{\lambda}_s, \mathbf{P}_s, \mathbf{b}_{t,s} \right), \tag{3}$$

where $m_{s,i} = 1$ indicates that site i in snapshot s is masked, and $n_{\mathrm{mask}} = \sum_{s=1}^{B} \sum_{i=1}^{N} m_{s,i}$ is the total number of masked outcomes in the mini-batch. Here, p_θ denotes the predicted conditional outcome distribution and θ denotes the model parameters. The loss is therefore averaged over all masked outcomes in the mini-batch, with unmasked positions excluded.

Model parameters are optimized using AdamW [41]. Experiment-specific learning rates, batch sizes, training durations, masking parameters, learning-rate schedules and checkpoint-selection criteria are reported in the Supplementary Methods.

Iterative measurement generation

Diffushadow generates measurement snapshots through iterative unmasking [14]. Given a Hamiltonian control parameter $\boldsymbol{\lambda}$ and a target system size N, we independently sample a Pauli measurement basis from $\{X, Y, Z\}$ at each site and initialize all measurement outcomes as [MASK]. The control parameter and measurement bases remain fixed throughout generation.

At iteration k, let $\mathcal{M}_k$ denote the set of masked sites and $\mathbf{b}^{(k)}$ the current outcome sequence. The model predicts logits $z_i^{(k)}(b)$ for every masked site, conditioned on $\boldsymbol{\lambda}$, $\mathbf{P}$ and $\mathbf{b}^{(k)}$. These logits are converted into outcome probabilities using a sampling temperature $\tau > 0$:

$$q_i^{(k)}(b) = \frac{\exp\left(z_i^{(k)}(b)/\tau \right)}{\sum_{b' \in \{0,1\}} \exp\left(z_i^{(k)}(b')/\tau \right)}, \qquad b \in \{0, 1\}. \tag{4}$$

We rank the remaining masked sites by their predictive entropy and reveal one site per iteration. Specifically, the selected site and its sampled outcome are

$$\begin{aligned} H_i^{(k)} &= -\sum_{b \in \{0,1\}} q_i^{(k)}(b) \log q_i^{(k)}(b), \\ i_k &= \arg\min_{i \in \mathcal{M}_k} H_i^{(k)}, \\ \mathbf{b}^{(k+1)}[i_k] &\sim \mathrm{Bernoulli}\left(q_{i_k}^{(k)}(1) \right). \end{aligned} \tag{5}$$

Table 1 Three-round recursive scale extrapolation. Generated training records are supplied by the immediately preceding model, and each new model is trained from an independent random initialization. The generation targets list the sizes used to advance the recursive procedure and its final large-system outputs.

Round	Exact training sizes	Generated training sizes (source)	Generation targets
1	$8, 10, 12, 14$	—	$18, 20, 22, 24$
2	$12, 14$	$18, 20, 22, 24$ (round 1)	$24, 32, 40, 48$
3	$12, 14$	$24, 32, 40, 48$ (round 2)	$64, 72, 96$

The sampled outcome is inserted into the sequence and held fixed in subsequent iterations. Predictions for the remaining masked sites are then recomputed using the updated measurement context. This procedure generates a complete N-site record in N denoising steps, with an unmasking order determined adaptively for each snapshot.

Repeating the procedure with independently sampled measurement bases and outcome draws produces an ensemble of generated snapshots. Sampling temperatures and ensemble sizes for each benchmark are specified in the Supplementary Methods. The one-shot and random-position decoding controls, together with the ShadowGPT comparison settings, are also described there.

Recursive scale extrapolation

We extend Diffushadow from exact measurement data at small system sizes to larger one-dimensional systems through successive training rounds. Each round uses an independently initialized model, trained from scratch with the masking objective described in Methods ("Training Diffushadow"). After training, the model generates measurement records at larger system sizes using the iterative sampling procedure described in Methods ("Iterative measurement generation"). These generated records are combined with retained exact data to form the training dataset for the next round.

The first-round model is trained on exact classical-shadow records for $N \in \{8, 10, 12, 14\}$ and generates training records for the next round at $N \in \{18, 20, 22, 24\}$. The second-round model is trained on these generated records together with exact records retained at $N \in \{12, 14\}$, and generates training records for the next round at $N \in \{24, 32, 40, 48\}$. The third-round model is trained on the second-round generated records and the same exact records at $N \in \{12, 14\}$, and generates records for $N \in \{64, 72, 96\}$. Thus, the largest generated system contains 96 spins, starting from exact training data containing at most 14 spins. Table 1 summarizes the data sources and generation targets.

The retained exact records provide a common reference at small system sizes during the later training rounds, while the generated records extend the range of sizes represented in training. Finite-size ensembles used for thermodynamic-limit inference can therefore originate from different models in this sequence.

For the two-dimensional TFI benchmark, we use direct size extrapolation with a single trained model. The model trained on lattices containing at most 16 spins generates measurement records on 6×6 and 7×7 lattices with its parameters held fixed. These tests evaluate size generalization of that model, whereas the one-dimensional thermodynamic-limit analysis uses the recursively trained model sequence.

Observable estimation and thermodynamic-limit inference

For each evaluated control parameter and system size, physical observables are estimated from ensembles of $M = 10{,}000$ generated classical-shadow snapshots. Binary measurement outcomes are mapped to Pauli eigenvalues as $s_{t,i} = 2b_{t,i} - 1$. We use standard inverse-channel classical-shadow estimators [12], with spatial averaging over lattice origins and median-of-means aggregation over $K = 10$ equally sized groups. The same estimators are applied to exact and generated measurement records, and numerical references are evaluated using the same observable definitions. Complete estimator formulas and evaluation grids are provided in the Supplementary Methods.

Throughout, X, Y and Z denote Pauli matrices. For the one-dimensional TFI model, we evaluate $C_{ZZ,N}(r) = N^{-1}\sum_i \langle Z_i Z_{i+r}\rangle$. For the J_1–J_2 model, we evaluate $C_{\sigma,N}(r) = N^{-1}\sum_i \langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+r}\rangle$ at $r = 1, 2$, obtained by summing the three component-resolved estimates. This convention gives $\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_j\rangle = 4\langle \mathbf{S}_i \cdot \mathbf{S}_j\rangle$ for spin operators $\mathbf{S} = \boldsymbol{\sigma}/2$. The J_1–J_2 iDMRG references are averaged over both origins of the two-site unit cell to match the spatially averaged finite-chain observables.

For the ANNNI model at fixed $\kappa = 0.4$, we evaluate the longitudinal structure factors at $q = \pi$ and $q = \pi/2$, using

$$S_{Z,N}(q) = \frac{1}{N}\sum_{i,j=1}^{N} \cos[q(i-j)]\langle Z_i Z_j\rangle = \sum_{r=0}^{N-1} \cos(qr) C_{ZZ,N}(r), \tag{6}$$

with $C_{ZZ,N}(0) = 1$. The iDMRG references use connected correlations, $C^{\mathrm{conn}}_{ZZ}(r) = C_{ZZ}(r) - \langle Z\rangle^2$, and are converted to the same two-sided Fourier normalization. A uniform disconnected contribution cancels at the nonzero commensurate momenta considered here, allowing the finite-ring shadow estimator to use the full correlations. For the two-dimensional TFI lattices, we evaluate spatially averaged ZZ correlations at displacements $(0, 1)$, $(1, 0)$ and $(0, 2)$ in row–column coordinates, corresponding to horizontal nearest neighbours, vertical nearest neighbours and horizontal next-nearest neighbours. These quantities are compared with numerical references at the corresponding finite lattice sizes.

Thermodynamic-limit inference is performed for the one-dimensional observables using an empirical linear fit in inverse system size,

$$O_N(\boldsymbol{\lambda}) = O_\infty(\boldsymbol{\lambda}) + \frac{a(\boldsymbol{\lambda})}{N}. \tag{7}$$

The coefficients are determined by unweighted least squares independently for each observable, distance or momentum, and control-parameter value. The intercept at $1/N = 0$ provides the thermodynamic-limit estimate O_∞.

For the TFI model, the $r = 1$ fits use $N \in \{10, 12, 16, 18, 20, 22, 24, 32, 36, 48, 64, 72, 96\}$, whereas the $r = 10$ fits use $N \in \{24, 32, 36, 48, 64, 72, 96\}$. Records at $N \in \{10, 12, 16, 18, 20, 22, 24\}$, $N \in \{32, 36, 48\}$ and $N \in \{64, 72, 96\}$ are generated by the first-, second- and third-round models, respectively. The J_1–J_2 fits at $r = 1, 2$ use $N \in \{16, 18, 20, 22, 24, 32, 36, 40, 64, 72\}$.

Records at $N \in \{16, 18, 20, 22, 24\}$, $N \in \{32, 36, 40\}$ and $N \in \{64, 72\}$ are generated by the first-, second- and third-round models, respectively. The ANNNI fits at both momenta use $N \in \{16, 20, 24, 32, 36, 40, 48, 64\}$. Records at $N \in \{16, 20, 24\}$, $N \in \{32, 36, 40\}$ and $N \in \{48, 64\}$ are generated by the first-, second- and third-round models, respectively.

Sampling uncertainty in the one-dimensional finite-size and extrapolated estimates is quantified by nonparametric snapshot bootstrap resampling [42]. At each system size and control-parameter value, complete measurement records are independently resampled with replacement to produce $B = 1000$ bootstrap ensembles, each containing M snapshots. Resampling complete records preserves the correlations among outcomes within each snapshot. For every replicate, the observables are recomputed using the same spatial averaging and median-of-means procedure, and the linear finite-size fit is repeated. Central extrapolated estimates are obtained by fitting the bootstrap-mean observable at each system size.

The reported 95% intervals are the 2.5th and 97.5th percentiles of the bootstrap distributions of the finite-size observables or fitted intercepts. Intervals are evaluated separately at each control-parameter value and therefore represent pointwise intervals. The trained models and fitting protocol are held fixed throughout resampling. These intervals quantify finite-record sampling uncertainty propagated through observable estimation and finite-size fitting; they do not include systematic generation bias, errors inherited through recursive training, or uncertainty in the empirical extrapolation form.

Conditional reconstruction of Rydberg measurements

Rydberg reconstruction uses occupation records $\mathbf{n} \in \{0, 1\}^N$ measured in the fixed Z basis, where $n_i = 1$ denotes a Rydberg excitation. Diffushadow is trained on ground-state measurement records for $N \in \{16, 24, 32\}$ at $R_b/a = 2.35$, with training detunings extending up to $d = \delta/\Omega = 5.0$. Reconstruction is evaluated on open chains of $N = 32$ atoms using held-out MPS-sampled records generated independently of the training records. We use $M_d = 1000$ test records per detuning at $d \in \{0.2, 1.8, 3.4, 5.0\}$, and $M_d = 500$ at $d \in \{4.0, 4.4, 4.8, 5.2, 5.6, 6.0\}$. The last three detunings extend beyond the training range. The complete training grid and numerical settings are specified in the Supplementary Methods.

For each complete test record s, we uniformly select $k = \max\{1, \mathrm{round}(fN)\}$ distinct sites without replacement and replace their occupation tokens with [MASK]. Thus, the nominal missing fraction $f = 0.3$ corresponds to $k = 10$ missing sites per record. Let $\mathcal{M}_s$ denote the missing-site set and $\mathcal{O}_s = \{1, \ldots, N\} \setminus \mathcal{M}_s$ the observed-site set. All observed occupations remain fixed during completion. Seeded masks are generated separately for each record and shared across the decoding modes, controls and Bayes-reference calculation.

Both decoding modes use the iterative procedure described in Methods ("Iterative measurement generation"), with all basis tokens fixed to Z. At each iteration, the remaining missing sites are ranked by predictive entropy, and the lowest-entropy site is revealed. Predictions are recomputed using the updated context, while previously

revealed outcomes remain fixed. Revealing one site per iteration requires k updates to complete each record.

For ensemble reconstruction, the occupation at the selected site is sampled from the model's predicted distribution at temperature $\tau = 1$. One completion is produced for each held-out record. We evaluate occupation densities and the connected density–density structure factor,

$$S(q) = \frac{1}{N} \sum_{i,j=1}^{N} \cos[q(i-j)] \left(\langle n_i n_j \rangle - \langle n_i \rangle \langle n_j \rangle \right), \tag{8}$$

where the averages are taken over the completed ensemble. These quantities are compared with the intact records and reference MPS. The unconditioned-fill control generates independent full configurations conditioned only on detuning and inserts their occupations at the missing sites, retaining the original observed occupations. This control tests the contribution of conditioning on the observed pattern in each record.

For individual-record reconstruction, the selected occupation is instead assigned its most probable value by a stepwise argmax. Accuracy is evaluated as the fraction of missing occupations matching the original held-out record, averaged over all test records and missing sites. The unconditional site-density baseline predicts $\widehat{n}_j = \mathbf{1}[\rho_j(d) > 1/2]$, where $\rho_j(d) = \langle n_j \rangle_{\psi_d}$ is the site-resolved occupation probability of the reference MPS. This baseline uses detuning-dependent density information without conditioning on the observed occupations of the individual record.

To determine the optimal expected per-site accuracy under the same observed information, we condition the reference MPS $|\psi_d\rangle$ on the occupations in $\mathcal{O}_s$. Contracting the resulting tensor-network environments gives the conditional marginal

$$p_{s,j} = \Pr_{\psi_d} \left(n_j = 1 \mid \mathbf{n}_{\mathcal{O}_s}^{(s)} \right), \qquad j \in \mathcal{M}_s. \tag{9}$$

The Bayes-optimal expected per-site accuracy is then

$$A_{\mathrm{Bayes}}(d) = \frac{1}{M_d k} \sum_{s=1}^{M_d} \sum_{j \in \mathcal{M}_s} \max(p_{s,j}, 1 - p_{s,j}). \tag{10}$$

This reference averages the optimal binary decision at each missing site under the reference MPS distribution. The conditional-marginal calculation is checked by verifying that, with no sites observed, it reproduces the MPS occupation profile. Reference-state settings, tensor-contraction details, and definitions of the blockade and multisite-correlation diagnostics are provided in the Supplementary Methods.

Attention analysis

We analyze attention maps from a fixed trained TFI checkpoint in evaluation mode. All measurement-outcome tokens are set to [MASK], while the Hamiltonian parameter and measurement bases remain available. For each system size, Pauli-basis

configurations are sampled independently and uniformly at each site, and the same configurations are reused across the scanned values of g. The checkpoint identifier, training sizes, number of basis configurations, random seeds and evaluation grids are specified in the Supplementary Methods.

Let $A_{qk}^{(\ell,h)}$ denote the attention weight from query token q to key token k in layer ℓ and head h. We use masked outcome tokens as queries and assign both P_i and b_i to lattice site i. For the set of site-indexed keys $\mathcal{S} = \{P_i, b_i\}_{i=1}^{N}$, the normalized attention-weighted site distance is

$$D^{(\ell,h)}(q) = \frac{\sum_{k\in\mathcal{S}} A_{qk}^{(\ell,h)} d_N(s(q), s(k))}{\lfloor N/2 \rfloor \sum_{k\in\mathcal{S}} A_{qk}^{(\ell,h)}}, \tag{11}$$

where $d_N(i,j) = \min\{|i-j|, N-|i-j|\}$ is the shortest periodic lattice distance. The denominator renormalizes attention over the basis and outcome keys and scales distances by the maximum periodic separation. The parameter token is excluded from this spatial measure. The mean distance reported in Fig. 2e is averaged over query sites, basis configurations, heads and layers.

To resolve outcome-directed attention by spatial separation, we define

$$K_{bb}^{(\ell,h)}(\delta; N, g) = \left\langle \sum_{j:\, \delta_N(i,j)=\delta} A_{b_i,b_j}^{(\ell,h)} \right\rangle_{i,\mathbf{P}}, \tag{12}$$

where $\delta_N(i,j)$ is the signed periodic representative of $j-i$ in $(-N/2, N/2]$. The average is taken over query sites and basis configurations, retaining a separate kernel for each layer–head pair. Absolute-distance kernels are obtained by combining signed displacements,

$$K_{bb,\mathrm{abs}}^{(\ell,h)}(r) = \sum_{\delta:\, |\delta|=r} K_{bb}^{(\ell,h)}(\delta). \tag{13}$$

This construction counts the self-site contribution and, for even N, the antipodal site once.

Each head's folded kernel is compared with the corresponding exact finite-size correlation profile $C_{ZZ,N}(r,g) = |\langle Z_i Z_{i+r}\rangle|$, with spatial averaging understood. The comparison set $\mathcal{R}_N$ contains the nonzero periodic distances available in both the kernel and the numerical reference. The illustrated $N = 14$ profile in Fig. 2f uses $r = 1, \ldots, 5$; the complete comparison ranges and reference-grid matching settings are reported in the Supplementary Methods. Each profile is divided by its sum over $\mathcal{R}_N$, giving $\widetilde{K}_{bb}^{(\ell,h)}(r)$ and $\widetilde{C}_{ZZ,N}(r)$. Their cosine similarity is

$$S_{\cos}^{(\ell,h)}(N,g) = \frac{\sum_{r\in\mathcal{R}_N} \widetilde{K}_{bb}^{(\ell,h)}(r)\widetilde{C}_{ZZ,N}(r)}{\sqrt{\sum_{r\in\mathcal{R}_N}\left[\widetilde{K}_{bb}^{(\ell,h)}(r)\right]^2}\sqrt{\sum_{r\in\mathcal{R}_N}\left[\widetilde{C}_{ZZ,N}(r)\right]^2}}. \tag{14}$$

The similarity is defined for profiles with nonzero norms.

The single-head example in Fig. 2f uses L5H4 at $N = 14$ and $g = 0.7$, selected by the largest cosine similarity at this setting. Layer and head labels follow the zero-based implementation indices. For Fig. 2g, cosine similarities are first calculated separately for every layer–head pair after averaging its kernel over queries and basis configurations. At each N and g, the central curve shows the median across these pairs, and shading spans their 25th to 75th percentiles. This interquartile range describes variation among heads.

Basis-directed kernels, across-parameter distance variations ΔD, spatial first moments, and the fixation and KL-divergence protocols for the causal-mask control are described in the Supplementary Methods.

Data availability

No external datasets were used in this study. The data associated with this article can be reproduced using https://github.com/Miamm0v0/Diffushadow.

Code availability

The source code associated with this article is openly available at https://github.com/Miamm0v0/Diffushadow.

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Author contributions

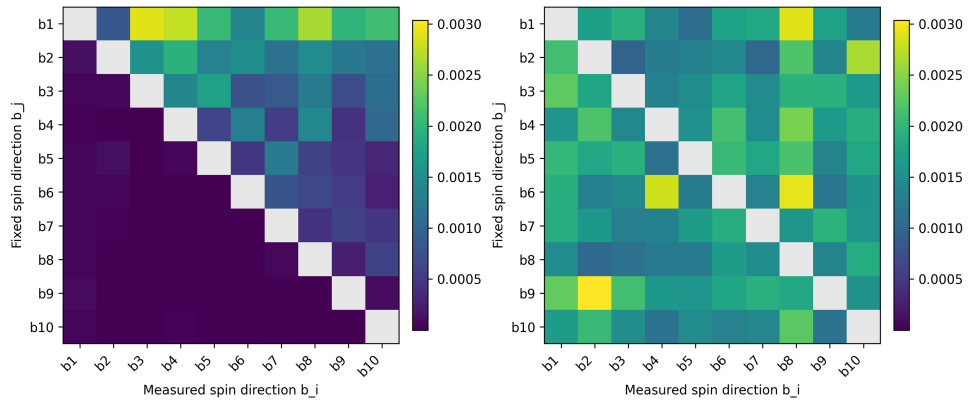
Y.L. conceived the study, developed and implemented the machine-learning framework, designed and conducted the numerical experiments, analysed the results and wrote the original manuscript. M.H. contributed to the theoretical and physical analysis, including the study of quantum many-body models, the analysis of physical observables and the interpretation of the numerical results. H.-K.T. supervised the research, led the Rydberg-chain investigations and contributed to the physical analysis and interpretation of the results. All authors discussed the findings, reviewed and revised the manuscript.

Competing interests

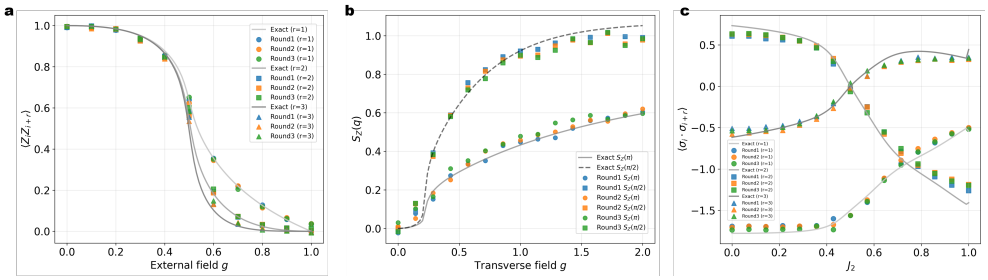
The authors declare no competing interests.

Extended Data Table 1 MSE between Model Predictions (Diffushadow, ShadowGPT) and Exact Values for the ground-state energy and three spin-spin correlations. ShGPT denotes ShadowGPT, Diffu-rdm denotes Diffushadow with randomly selected decoding positions, Diffu-s1 denotes Diffushadow decoding in one step, Diffu denotes the full Diffushadow decoding procedure.

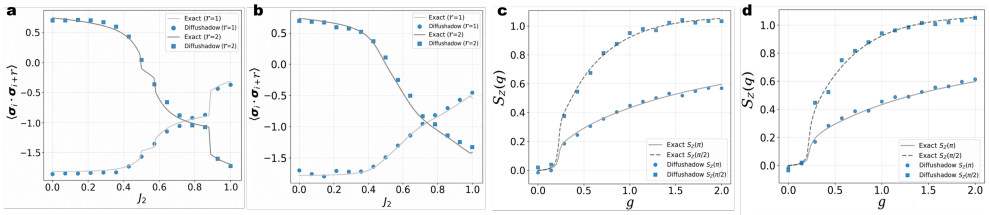
	Model	E_0	$\langle Z_i Z_{i+1} \rangle$	$\langle Z_i Z_{i+3} \rangle$	$\langle Z_i Z_{i+5} \rangle$
$g = 0.1$	ShGPT	3.63×10^{-4}	2.64×10^{-8}	6.51×10^{-6}	$\mathbf{9.00\times10^{-6}}$
	Diffu-rdm	2.34×10^{-3}	2.63×10^{-8}	9.00×10^{-6}	1.30×10^{-5}
	Diffu-s1	7.99×10^{1}	1.01×10^{0}	1.00×10^{0}	1.02×10^{0}
	Diffu	$\mathbf{2.77\times10^{-4}}$	$\mathbf{2.47\times10^{-8}}$	$\mathbf{1.44\times10^{-6}}$	1.10×10^{-5}
$g = 0.5$	ShGPT	1.35×10^{-2}	6.30×10^{-4}	1.09×10^{-3}	4.80×10^{-4}
	Diffu-rdm	7.59×10^{-3}	1.97×10^{-4}	9.42×10^{-4}	6.46×10^{-4}
	Diffu-s1	1.00×10^{1}	4.10×10^{-1}	2.58×10^{-1}	2.44×10^{-1}
	Diffu	$\mathbf{7.13\times10^{-3}}$	$\mathbf{1.61\times10^{-4}}$	$\mathbf{4.00\times10^{-4}}$	$\mathbf{2.12\times10^{-4}}$
$g = 0.9$	ShGPT	6.79×10^{-4}	6.27×10^{-4}	2.28×10^{-5}	5.20×10^{-5}
	Diffu-rdm	5.05×10^{-4}	1.26×10^{-4}	1.07×10^{-4}	1.10×10^{-4}
	Diffu-s1	1.47×10^{-3}	4.11×10^{-3}	2.58×10^{-5}	9.82×10^{-5}
	Diffu	$\mathbf{4.98\times10^{-4}}$	$\mathbf{5.39\times10^{-5}}$	$\mathbf{5.15\times10^{-6}}$	$\mathbf{3.70\times10^{-7}}$



Extended Data Fig. 1 KL divergence of the predicted distributions of the remaining qubits after fixing the measurement outcome of a selected qubit. (a) The result of Diffushadow with causal attention. (b) The result of Diffushadow with full-attention. Full attention allows the perturbation to influence both earlier and later sequence positions, whereas causal attention primarily propagates the perturbation to later positions.



Extended Data Fig. 2 Consistency of observable estimates across recursive training rounds. Observable estimates from Diffushadow models obtained in three recursive training rounds are compared at the common system size $N = 24$. **a,** Longitudinal spin correlations $\langle Z_i Z_{i+r} \rangle$ at $r = 1, 2, 3$ in the transverse-field Ising (TFI) chain. **b,** Longitudinal structure factors $S_Z(\pi)$ and $S_Z(\pi/2)$ in the ANNNI chain. **c,** Spin-dot correlations $\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+r} \rangle$ at $r = 1, 2, 3$ in the J_1–J_2 chain. Solid or dashed curves denote finite-size exact diagonalization (ED) references, while markers denote estimates from Diffushadow-generated classical-shadow measurements. The observable estimates remain broadly consistent across the three recursive rounds, indicating limited drift in the tested observables at this system size.



Extended Data Fig. 3 Finite-size observable validation for the J_1–J_2 and ANNNI chains. **a,b,** Spin-dot correlations $\langle \boldsymbol{\sigma}_i \cdot \boldsymbol{\sigma}_{i+r} \rangle$ at separations $r = 1$ (circles) and $r = 2$ (squares) for the J_1–J_2 model with $J_1 = 1$, shown as functions of J_2 for $N = 12$ (**a**) and $N = 24$ (**b**). **c,d,** Longitudinal structure factors $S_Z(q)$ at $q = \pi$ (circles) and $q = \pi/2$ (squares) for the ANNNI model at fixed $\kappa = 0.4$, shown as functions of g for $N = 12$ (**c**) and $N = 24$ (**d**). Blue markers denote estimates from Diffushadow-generated classical-shadow measurements, and grey lines denote finite-size exact diagonalization (ED) references. The size $N = 12$ is included in the initial training set, whereas $N = 24$ is absent from it.